

updated: 4/26

# Curriculum Vitae: Chad McKell

## ABOUT

|           |                                                                                                                                                                                                                                |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title     | Research Engineer   Acoustics, Audio & Applied Math                                                                                                                                                                            |
| Company   | Sigma Connectivity, on assignment at Meta                                                                                                                                                                                      |
| Address   | 10301 Willows Road NE, Redmond, WA 98052                                                                                                                                                                                       |
| Email     | cmckell@meta.com                                                                                                                                                                                                               |
| Website   | chadmckell.com                                                                                                                                                                                                                 |
| Languages | English (native), Spanish (B2), French (A2)                                                                                                                                                                                    |
| Code      | Python, C++/CUDA, MATLAB, LaTeX                                                                                                                                                                                                |
| Research  | <i>acoustics</i> : numerical wave simulation, exterior wave problems<br><i>audio</i> : personalized spatial audio, audio ML/AI, speech enhancement<br><i>applied math</i> : conformality, differential forms, variational math |

## EDUCATION

|             |                                                                                                                                                                                                                                                                                |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (exp. 2026) | <b>University of California San Diego</b> , Ph.D. in Computer Music<br>Research area: Computational Acoustics and Applied Mathematics<br>Dissertation: <i>Conformal Symmetry for Exterior Wave Problems</i><br>Co-advisors: Albert Chern (CSE), Miller Puckette (Music; IRCAM) |
| 2017        | <b>University of Edinburgh</b> , M.S. in Acoustics and Music Technology                                                                                                                                                                                                        |
| 2015        | <b>Wake Forest University</b> , M.S. in Physics                                                                                                                                                                                                                                |
| 2009        | <b>Brigham Young University</b> , B.S. in Biophysics                                                                                                                                                                                                                           |

## EMPLOYMENT

|            |                                                                                    |
|------------|------------------------------------------------------------------------------------|
| 9/24–      | <b>Sigma Connectivity, on assignment at Meta</b> , Research Engineer               |
| 3/24–6/24  | <b>University of California San Diego</b> , Teaching Assistant in Computer Science |
| 6/23–2/24  | <b>Meta, Reality Labs Research</b> , Research Scientist Intern                     |
| 8/21–3/22  | <b>Meta, Reality Labs Research</b> , Research Intern                               |
| 7/18–7/19  | <b>Applied Research in Acoustics</b> , R&D Scientist                               |
| 4/17–9/17  | <b>Lofelt</b> , Research Consultant – Acquired by Meta in 2022                     |
| 10/14–8/16 | <b>J.P. Morgan/Neovest (via ConsultNet)</b> , SDET                                 |
| 9/09–9/11  | <b>Wake Forest University</b> , Teaching Assistant in Physics                      |

## RESEARCH ACTIVITIES

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9/24–     | <b>Sigma Connectivity, on assignment at Meta</b> , Research Engineer<br>Redmond, Washington. Research area: <i>spatial audio</i> . Audio research for Reality Labs<br>Research at Meta. Supervisor: Sebastian Prepelitã. Team Lead: Ravish Mehra.                                                                                                                                                                                                                                                                                                                 |
| 9/19–9/24 | <b>University of California San Diego</b> , Ph.D. Student<br>La Jolla, California. Center for Visual Computing; Department of Music. Research<br>area: <i>computational acoustics</i> , <i>applied mathematics</i> . Mathematical and numerical<br>modeling for infinite-domain wave simulations, with applications in virtual acoustics<br>and related fields. Committee: Albert Chern (co-chair, CSE), Miller Puckette (co-chair,<br>Music; IRCAM), Melvin Leok (Mathematics), Stefan Bilbao (IRCAM), Sebastian<br>Prepelitã (Meta), Shahrokh Yadegari (Music). |

## RESEARCH ACTIVITIES CONT.

- 6/23–2/24      **Meta, Reality Labs Research**, Research Scientist Intern  
Redmond, Washington. Research area: *spatial audio*. Audio research for Reality Labs Research at Meta. Included part-time research period (12/23–2/24). Supervisor: Sebastian Prepelitǎ. Team Lead: Ravish Mehra.
- 8/21–3/22      **Meta, Reality Labs Research**, Research Intern  
La Jolla, California. Included part-time research period (1/22–3/22). Research description: see above.
- 7/18–7/19      **Applied Research in Acoustics**, R&D Scientist  
Culpeper, Virginia. Research area: *underwater acoustic sensing, acoustic beamforming*. Algorithm development for underwater acoustic recognition, with applications in naval sonar systems. Team Lead: Jonathan Botts.
- 1/17–8/17      **University of Edinburgh**, Master’s Student  
Edinburgh, Scotland. Acoustics & Audio Group, Reid School of Music. Research area: *computational physics*. Numerical algorithm development for audio-haptic devices, including the Razer Nari Ultimate headphones. Partially funded by Lofelt, a German haptics startup acquired by Meta in 2022. Advisor: Stefan Bilbao.
- 1/10–9/13      **Wake Forest University**, Master’s Student  
Winston-Salem, North Carolina. Department of Physics. Research area: *optical trapping, fluid dynamics*. Design, construction, and analysis of standing-wave optical traps for Brownian particle tracking. Published work in the Journal of the Optical Society of America B. Advisor: Keith Bonin.
- 8/07–8/09      **Brigham Young University**, Undergraduate Student  
Provo, Utah. Department of Physiology and Developmental Biology. Research area: *biophysics*. AFM imaging of artificial lipid bilayers. Advisor: David Busath.

## TEACHING EXPERIENCE

### As Instructor

**UNCSA**  
SCI 1100      General Physics. Fall 2012 (1 term).

### As TA

**UCSD**  
CSE 291      Physics Simulation. Spring 2024 (1 term).  
MUS 6      Electronic Music. Fall 2020 (1 term).  
MUS 171      Computer Music I. Winter 2022 (1 term).  
MUS 172      Computer Music II. 2021–2022 (2 terms).

**WFU**  
PHY 113      General Physics I (Mechanics). 2009–2011 (4 terms).  
PHY 114      General Physics II (E&M). Fall 2010 (1 term).

## PUBLICATIONS

### Manuscripts in Progress

- (1) **C. McKell**, M. Nabizadeh, S. Wang, and A. Chern, “Wave simulations in infinite spacetime”.

*Simulating wave propagation on an infinite domain has been a long-standing computational challenge. Conventional approaches to this problem only produce wave simulations on a small subset of the infinite domain. Using the fact that wave propagation on an infinite Minkowski spacetime is equivalent to wave propagation on a bounded Minkowski spacetime under a Kelvin-like transformation, we simulate wave propagation on the entire infinite domain using a finite discretization of the bounded domain with no additional loss of accuracy from the transformation.*

- (2) **C. McKell**, *Conformal Symmetry for Exterior Wave Problems*, Ph.D. Dissertation, University of California San Diego. Advisors: Albert Chern and Miller Puckette.

*This work presents novel geometric methods for handling open exterior boundaries in wave simulations. First, I discuss extensions to the reflectionless discrete perfectly matched layer for the wave and Helmholtz equations. Then, I demonstrate that the conformal invariance of the wave equation under a Kelvin transform in Minkowski spacetime allows one to convert an infinite domain problem into a bounded domain problem that can be solved using standard numerical methods with no additional loss of accuracy introduced by the transform. I conclude by discussing parallel computing strategies for the geometric boundary models, applications in architectural acoustics and binaural audio, and future work in obstacle boundary flattening.*

### Journal Articles

- (3) **C. McKell** and K. Bonin, “Optical corral using a standing-wave Bessel beam,” *Journal of the Optical Society of America B*, Vol. 35, No. 8, 1910–1920, 2018.

### Conference Proceedings

- (4) **C. McKell**, “Sonification of optically-ordered Brownian motion,” In Proceedings of the International Computer Music Conference (ICMC), Utrecht, Netherlands, September 2016.

### Master’s Theses

- (5) **C. McKell**, *Real-Time Physical Modeling for Haptic Feedback Rendering*, Master’s Thesis, University of Edinburgh, Acoustics and Audio Group, 2017. Advisor: Stefan Bilbao.
- (6) **C. McKell**, *Confinement and Tracking of Brownian Particles in a Bessel Beam Standing Wave*, Master’s Thesis, Wake Forest University, Department of Physics, 2015. Advisor: Keith Bonin.